

LivingLink

Open-source support across the entire living kidney donor journey

FUNCTIONAL BETA • PILOT READY

THE PROBLEM & SOLUTION

THE CORE PROBLEM

Willing donors face disconnected information, financial uncertainty, isolation, evaluation delays, and weak long-term follow-up. Each handoff creates another opportunity to withdraw.

OUR NOVEL APPROACH

One donor-centered platform combines five modules: ReadyCheck, DonorShield, MentorMatch, CenterFlow, and LifeAfter. Five public tools remove the sign-up wall; role-based workflows connect donor and center tasks.

OUR TEAM & CAPABILITY

- Project lead: Richard Horsley, submission lead and prototype owner.
- Repository capability: Next.js/TypeScript, Prisma/PostgreSQL, donor workflows, FHIR integration prototypes, security and accessibility planning.
- Pilot partners: two transplant centers to be recruited; no institutional pilot partner is represented as confirmed.
- Required advisors: prior donor, coordinator, transplant nephrologist, interoperability, security, privacy, and accessibility reviewers.

MEASURABLE VALUE • 12-MONTH PILOT TARGETS (NOT YET CLINICALLY VALIDATED)

26% → 75%

NLDAC COMPLETION

-20%

DONOR WITHDRAWAL

-6 weeks

EVALUATION DURATION

59% → 85%

TWO-YEAR FOLLOW-UP

WHY IT CAN SCALE

- MIT-licensed code and a single web application reduce deployment complexity and vendor lock-in.
- Public education works without an account; authenticated workflows add role, center, consent, and patient-context boundaries.
- No specialized hardware. Proposed pilots validate workflow fit, safety, accessibility, interoperability, and outcomes before clinical use.

Evidence of Efficacy & Prototype Showcase

EVIDENCE OF POTENTIAL EFFICACY

EVIDENCE-BASED DESIGN

- Peer education is backed by a submission-cited randomized trial reporting 23% higher donation intent among ambivalent candidates.
- DonorShield puts wage estimates and NLDAC guidance before clinical steps, addressing financial uncertainty early.
- CenterFlow flags evaluations after 14 days without progress; LifeAfter pairs scheduled check-ins with PHQ-2 escalation at scores ≥ 3 .
- Four pilot outcomes are pre-specified: NLDAC completion, withdrawal, evaluation duration, and two-year follow-up.

FUNCTIONAL BETA SURFACES

- 5 public experiences: eligibility pre-screen, ripple calculator, waitlist map, stories, AI conversation practice.
- 5 authenticated modules plus donor, coordinator, clinician, patient, and administrator views.
- Prisma-backed APIs for goals, expenses, NLDAC, mentoring, evaluations, check-ins, consent, privacy, and alerts.

No pilot outcome or clinical efficacy is claimed yet.

TECHNICAL INNOVATION & AI USE

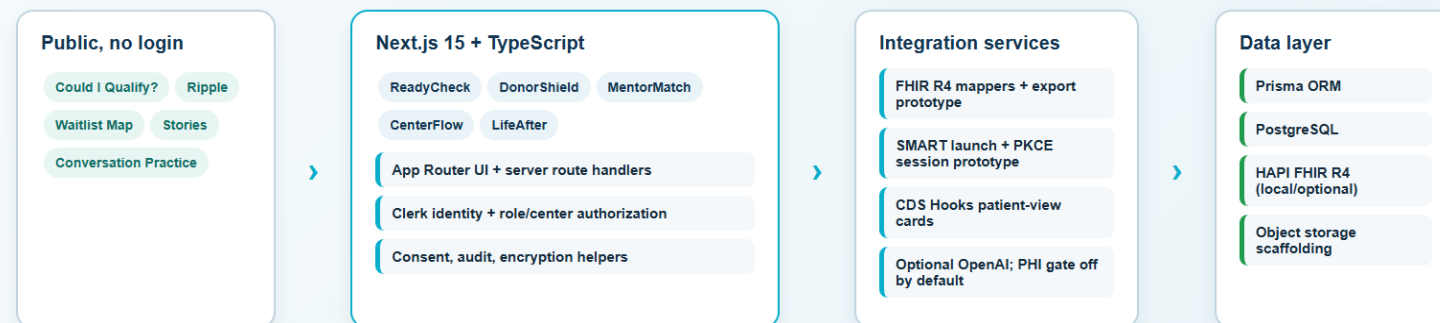
FHIR R4 mappings, SMART launch with PKCE, authenticated CDS Hooks cards, and pseudonymized export are implemented as prototypes. Optional GPT-4o supports public role-play and contextual assistance; authenticated health-data processing is configuration-gated and off by default.

SYSTEM ARCHITECTURE / UX PROTOTYPE

LivingLink Prototype Architecture

One Next.js application connects public donor tools, role-based workflows, optional services, and PostgreSQL.

Vendor sandbox + conformance validation pending



FHIR model: Patient, Observation, Goal, Coverage, Task, Communication, CarePlan, QuestionnaireResponse

Safety boundary: prototype only; no real PHI or clinical use until security, accessibility, interoperability, and clinical reviews are complete

Path to Implementation & Compliance

TECHNICAL FEASIBILITY

- Single Next.js 15 / TypeScript application with server route handlers.
- Prisma + PostgreSQL data layer; Clerk integration and role/center authorization helpers.
- Local HAPI FHIR R4 plus FHIR mapping, export, SMART, and CDS Hooks prototypes.
- Automated unit, integration, end-to-end, and limited axe-core accessibility test assets.

INTEROPERABILITY

- FHIR R4 resources: Patient, Observation, Goal, Coverage, Communication, Task, CarePlan, and QuestionnaireResponse.
- SMART launch and patient-view CDS Hooks preserve issuer, center, patient mapping, consent, and minimum-necessary boundaries.
- Epic and Oracle Health registration, sandbox testing, conformance validation, and production approval remain pilot milestones.

DEVELOPMENT ROADMAP

M1–M2

Assure

Assign clinical/security owners; risk analysis, penetration test, data governance, manual accessibility audit.

M3–M4

Validate

Center 1 workflow pilot; FHIR validator, usability, keyboard/screen-reader, and safety testing.

M4–M6

Integrate

Approved Epic/Oracle Health sandboxes; SMART/CDS contracts, identity, replay, token, and tenant tests.

M6–M12

Measure & Scale

Center 2 pilot; Spanish localization; outcome analysis; open-source release and deployment playbook.

COMPLIANCE PLAN

FIPS 199

Provisional HIGH impact under the high-water-mark rule; system boundary and owner approval pending.

SECURITY

Selected AES-256-GCM fields, consent, application audit logging, and RBAC exist; key management, complete coverage, monitoring, BAAs, and external testing are required before PHI.

ACCESSIBILITY

Section 508 / WCAG 2.1 AA is a target. Complete authenticated axe coverage, keyboard and assistive-technology testing, chart data tables, PDF remediation, and VPAT/ACR assessment.

PILOT GATE: no clinical use or real PHI until security, privacy, clinical, accessibility, interoperability, and partner approvals are evidenced.